

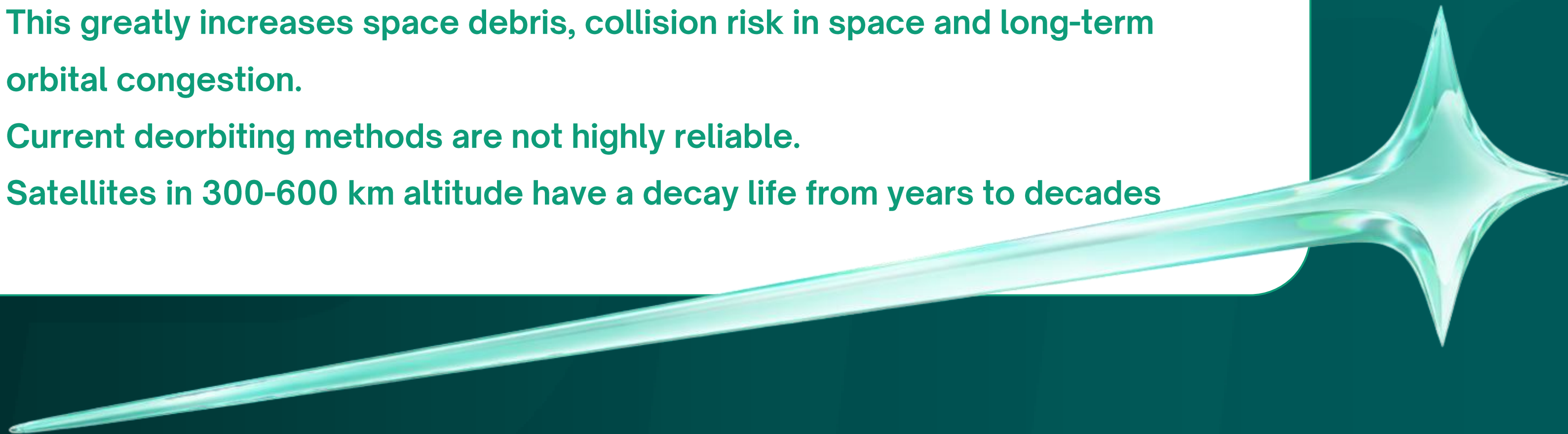
Ad Astra

Hybrid Drag
Sails(HDS)



Inefficient deorbiting methods

- Satellites in Low Earth orbit remain in orbit for years to decade after end-of-life due to very thin atmosphere drag.
- This greatly increases space debris, collision risk in space and long-term orbital congestion.
- Current deorbiting methods are not highly reliable.
- Satellites in 300-600 km altitude have a decay life from years to decades



Proposed Solution

Hybrid resilient drag sail
system.

- A drag sail system that accelerates orbital decay using atmospheric drag.
- It is more resilient against micrometeoroid and debris impacts.
- Keeps stability within control during deployment.
- Using passive heating to deploy.



System Architecture & Technical Design

Drag sail structure:

- Large ultra-thin membrane (~10–25 m scale depending on satellite class)
- Reinforced grid using:
 - Kevlar fibers
 - Nextel ceramic fibers
- Multi-layer rip-stop architecture (like parachutes but space-rated)

Passive Attitude Behavior:

- Aerodynamic drag torque stabilizes orientation
- Designed center-of-pressure vs center-of-mass offset for passive alignment

Shape Memory Deployment System:

- Uses Nitinol (Ni-Ti Shape Memory Alloy) booms
- Stored compactly during launch
- Deployed after EOL:
- Sunlight heats SMA → returns to programmed deployed shape
- No reliance on batteries or active control

Sail Geometry (Stability Concept):

- Modified pyramidal / conical sail geometry
- Purpose:
 - Reduce tumbling
 - Maintain aerodynamic stability in rarefied LEO atmosphere
- Self-align with velocity vector to maximize drag

Technical Implementation

Deployment Sequence:

- End-of-life trigger (timer or ground command failure fallback)
- Release of SMA booms
- Solar heating activates full extension
- Sail locks into pyramid-like geometry
- Passive drag increases orbital decay rate

Survivability Mechanism:

- Rip-stop fiber grid prevents propagation of punctures
- Localized damage does not collapse full structure
- Redundant load paths maintain structural integrity



Expected Results

Based on existing drag sail missions like:

- NASA NanoSail-D2
- The Planetary Society LightSail
- ESA ADEO systems

Expected improvements:

-Deorbit time reduction:

- From 20–50 years → 1–5 years (depending on altitude and mass)

-Increased survivability against small debris impacts

-Reduced reliance on onboard power after EOL

Impact

Major reduction in
long-term space
debris

Supports sustainable
space operations

Improves compliance
with “25-year deorbit
rule” (international
guideline for LEO
disposal)

Low-cost passive
deorbit solution

Applicable to
CubeSats and small
satellites (fast-
growing market
segment)

SDG Alignment

SDG 9 – Industry, Innovation & Infrastructure

- Improves space infrastructure sustainability

SDG 11 – Sustainable Cities & Communities

- Protects Earth orbital environment (long-term infrastructure safety)

SDG 12 – Responsible Consumption & Production

- Reduces orbital waste

SDG 13 – Climate Action

- Indirect contribution via sustainable Earth-space systems

SDG 17 – Partnerships

- Applicable to international space agencies and private sector collaboration

9 INDUSTRY, INNOVATION
AND INFRASTRUCTURE



11 SUSTAINABLE CITIES
AND COMMUNITIES



12 RESPONSIBLE
CONSUMPTION
AND PRODUCTION



13 CLIMATE
ACTION



17 PARTNERSHIPS
FOR THE GOALS



Scalability & Future Work

Stability Enhancements

- Active damping via passive geometry (no actuators)
- Aerodynamic self-stabilization in LEO flow regime
- Multi-point tether reinforcement to reduce oscillation modes

Future Improvements

- Adaptive sail geometry (deployable variable shape)
- Smart materials that change reflectivity + drag ratio
- Integration with AI-based orbital decay prediction systems
- Hybrid systems combining drag sail + electrodynamic tether



Challenges & Limitations

Engineering Challenges

- Very low atmospheric density → unpredictable drag variations
- Structural fatigue due to thermal cycling (-150°C to $+120^{\circ}\text{C}$)
- Micrometeoroid and orbital debris impacts
- Deployment failure risk in SMA systems

Physical Limitations

- Ineffective above ~1000 km (too little atmosphere)
- Large surface area required → packaging constraints
- Attitude control cannot be fully guaranteed in all conditions

Operational Risks

- Satellite must survive until deployment trigger
- Interaction with existing spacecraft in crowded orbits

Conclusion

This proposed drag sail system offers a resilient and passive solution for accelerating satellite deorbiting while addressing key limitations of current designs, such as structural fragility, instability, and power dependency. By integrating shape-memory deployment, debris-tolerant materials, and a stable pyramidal geometry, it enhances the sustainability and safety of Low Earth Orbit environments.